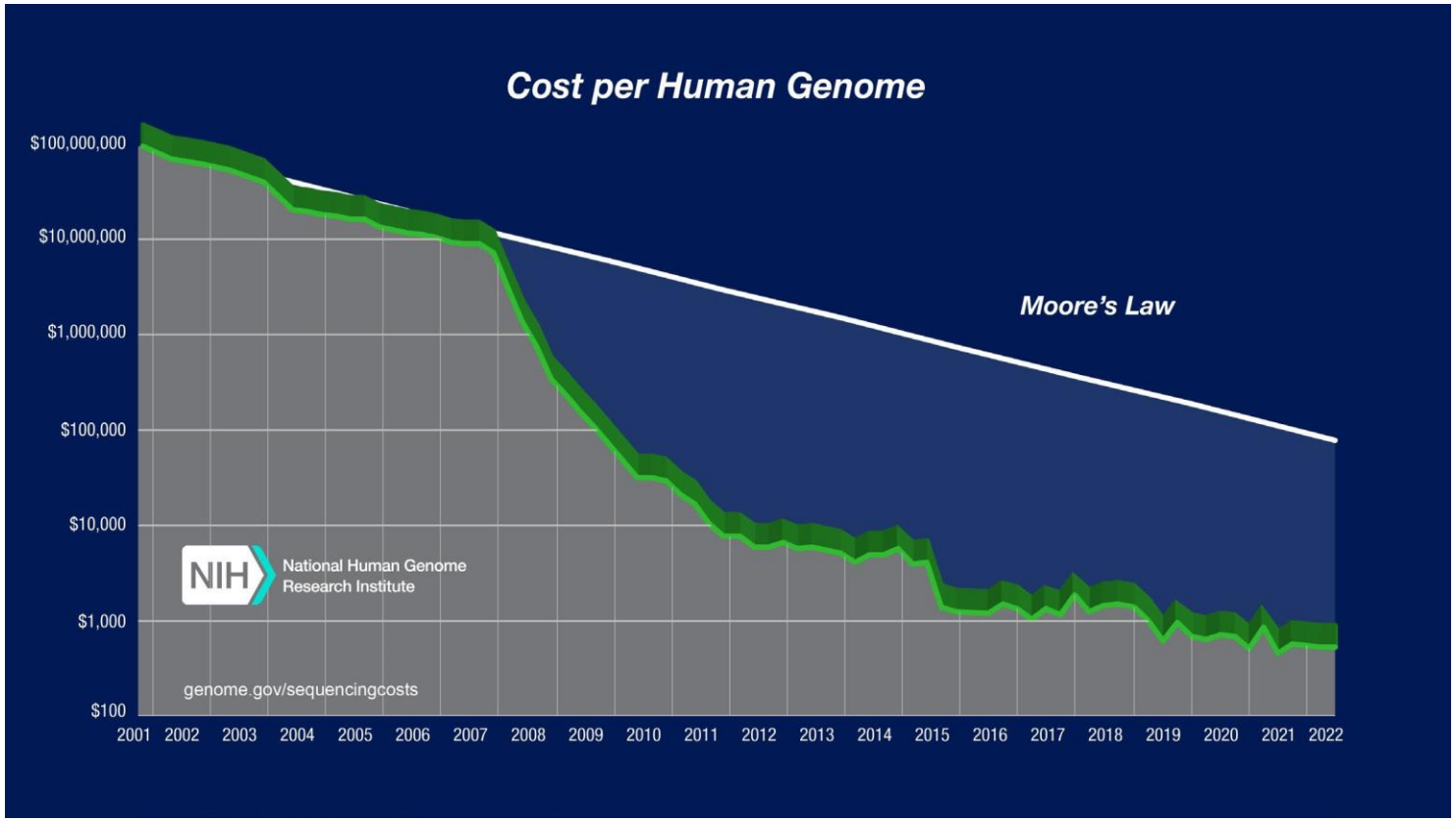
Spatial Transcriptomics Race

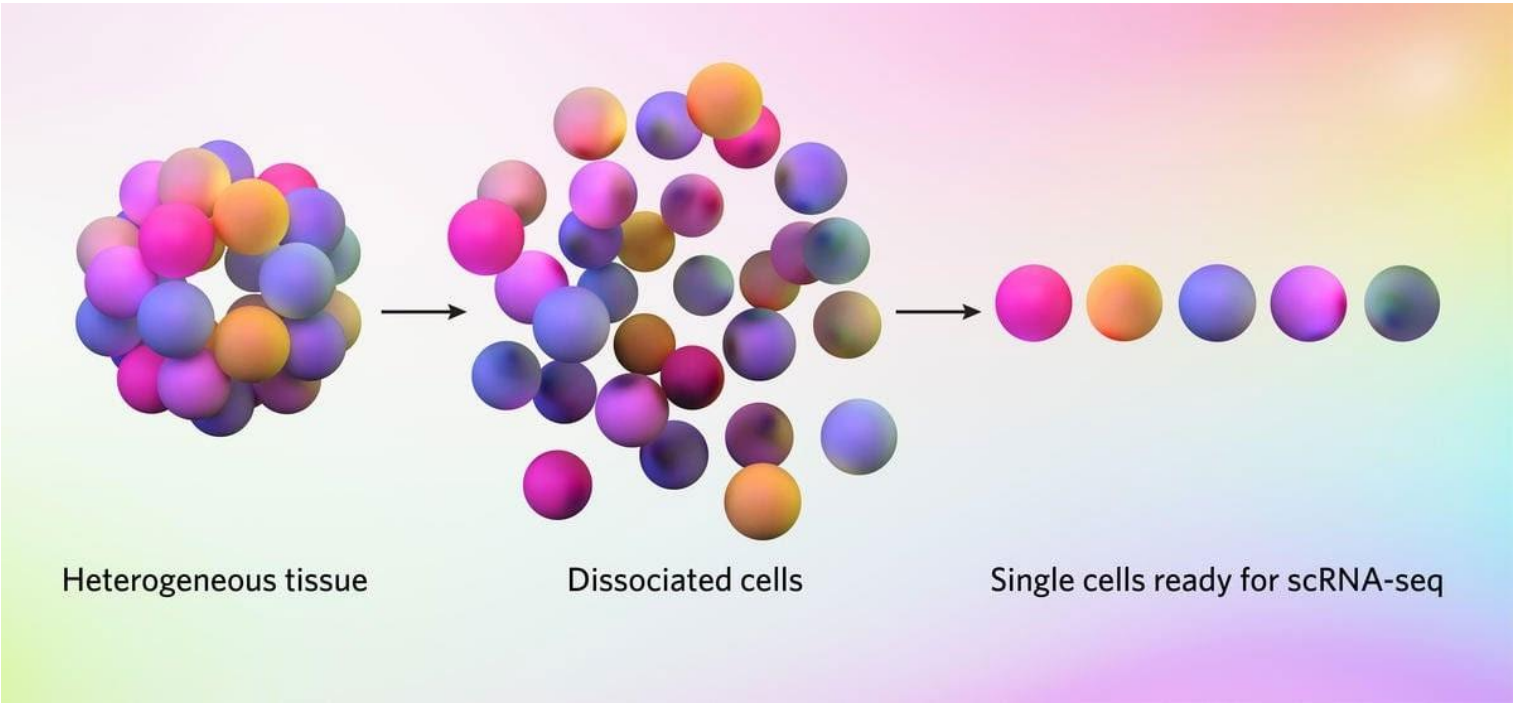
Recent Genomics Technology Trends

- More data at lower cost: sequencing technologies are still rapidly evolving



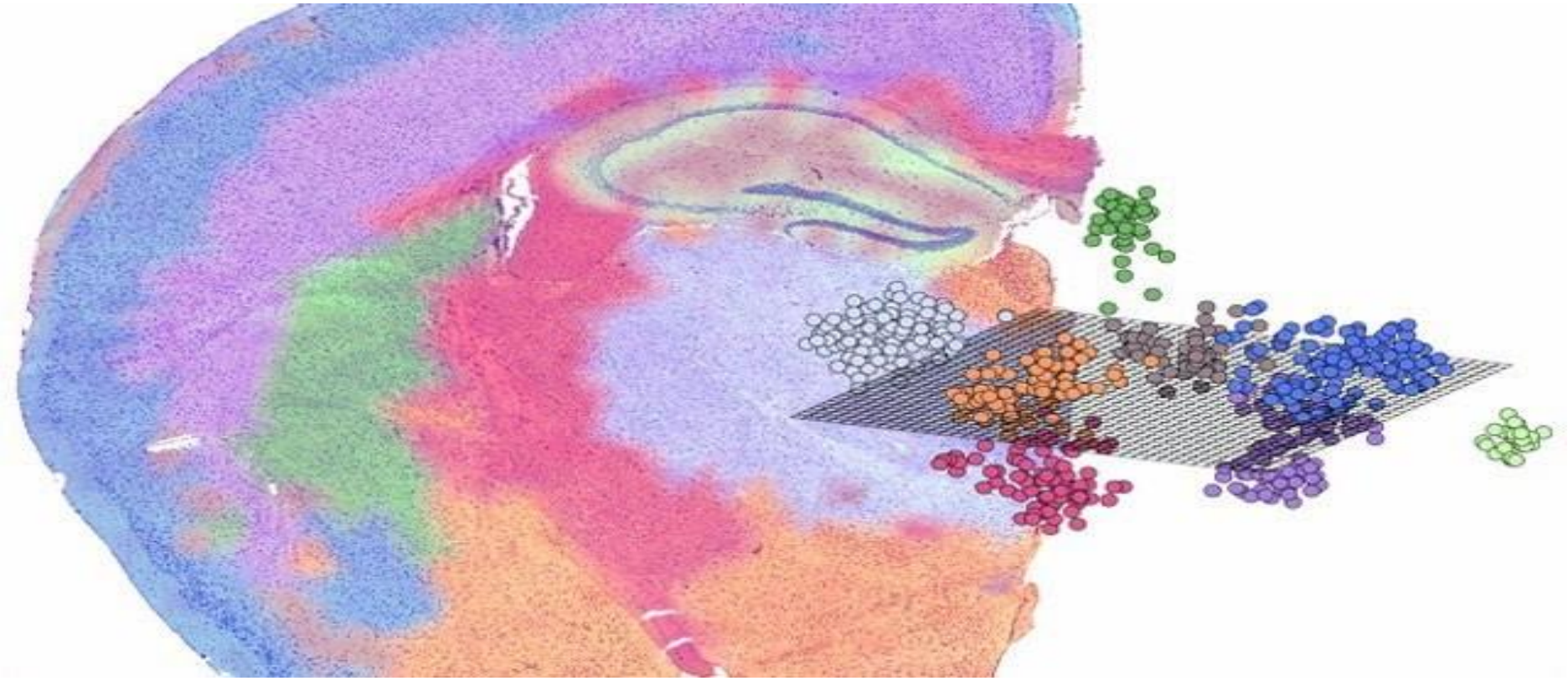
Recent Genomics Technology Trends

- Higher resolution: from bulk tissue/cell population to individual cells



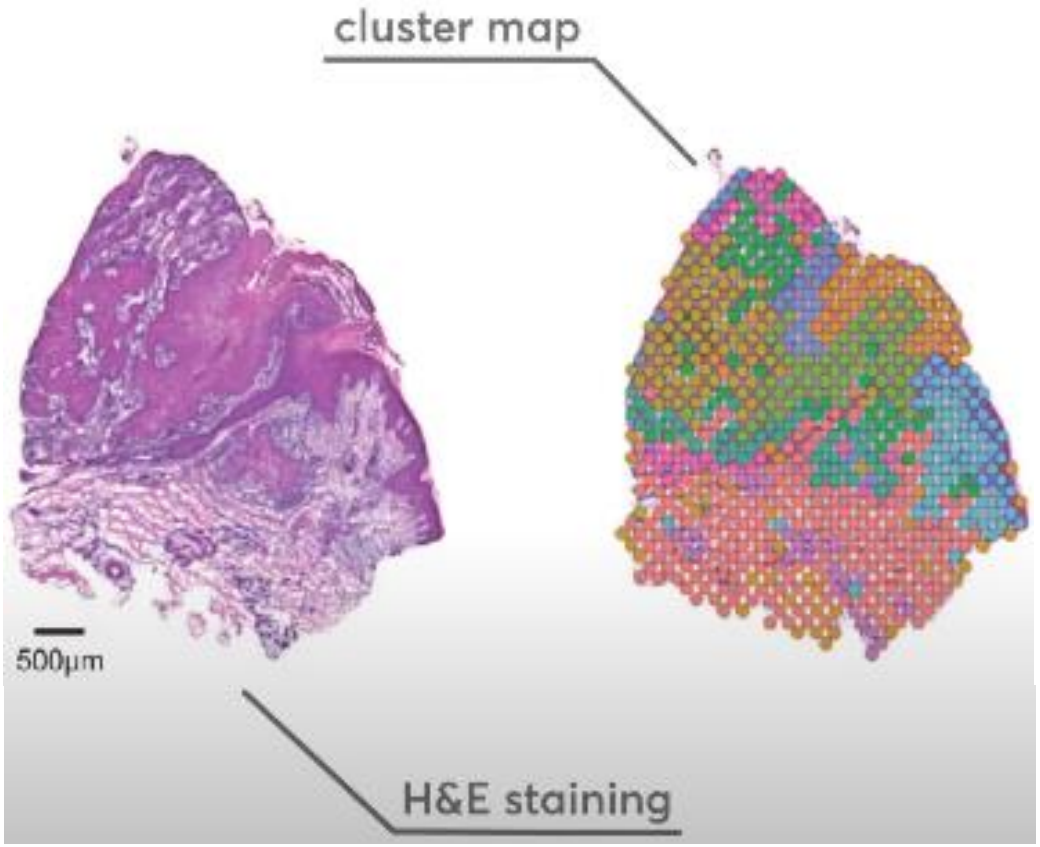
Recent Genomics Technology Trends

- Race into space: Spatial context matters



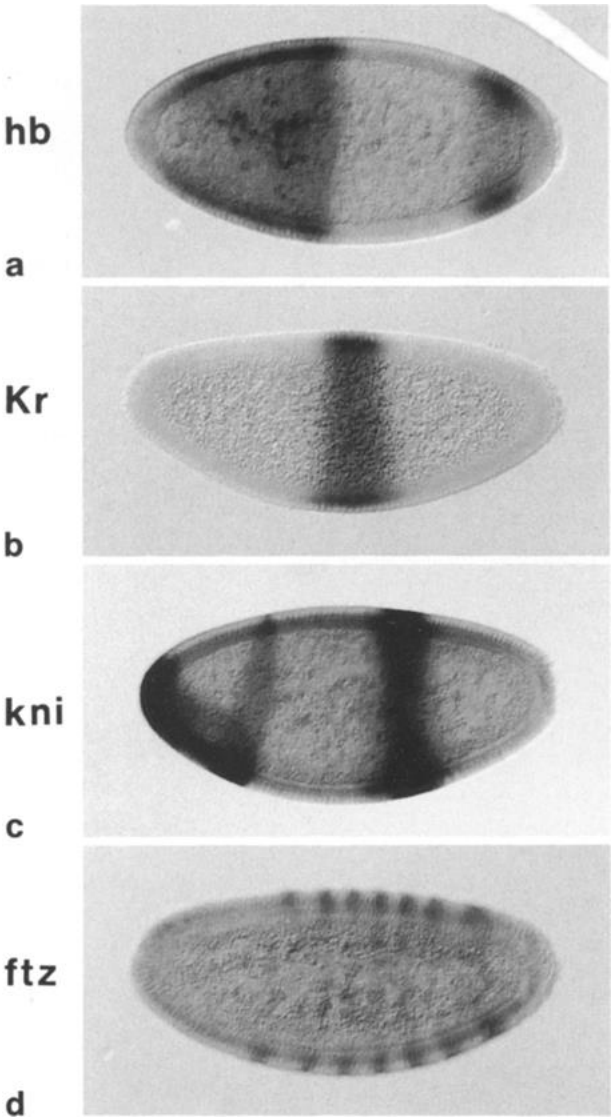
Recent Genomics Technology Trends

- Race into space: Spatial context matters



Spatial Transcriptomics

- Not a brand new technology: started from 1970s, but at a much lower throughput – one gene at a time
- The current surge in spatial transcriptomics is due to the demand to simultaneously quantify a large number of genes, from hundreds, thousands, to the entire genome



Localization of specific RNAs in Drosophila embryos.

Chromosoma (Berl) (1989) 98:81-85

Key Spatial Transcriptomics Technologies

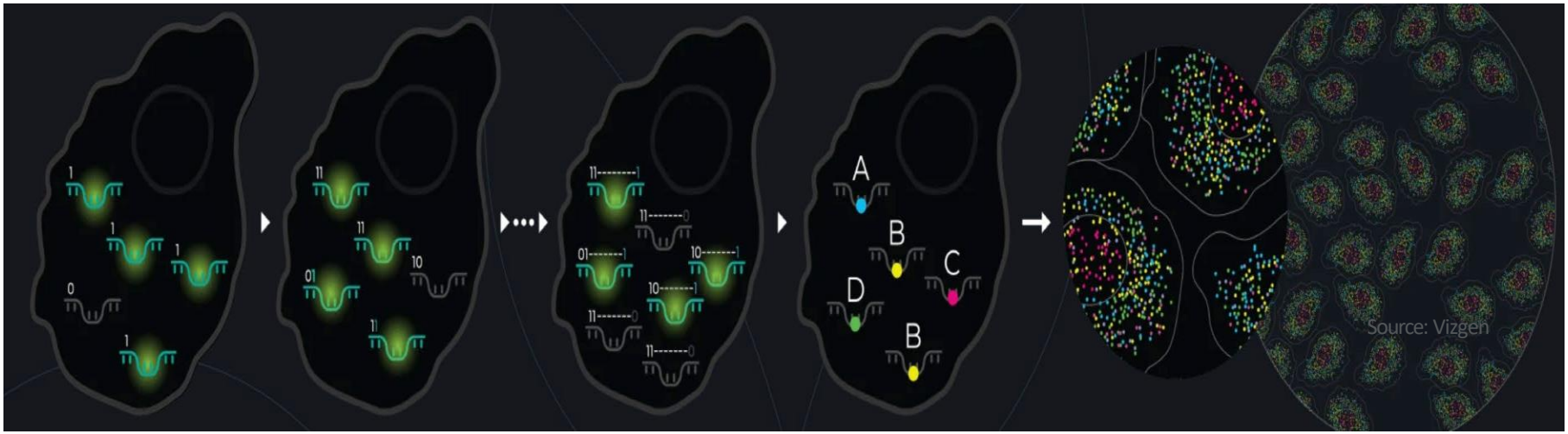
Three major approaches

- In-Situ Hybridization (ISH)
- In-Situ Sequencing (ISS)
- In-Situ Capturing (ISC)

Key Spatial Transcriptomics Technologies

In-Situ Hybridization (ISH)

- Highly multiplexed single molecule fluorescent in situ hybridization (smFISH)
  - Augmentation of the traditional in situ hybridization approach
  - Example: MERFISH (Vizgen)

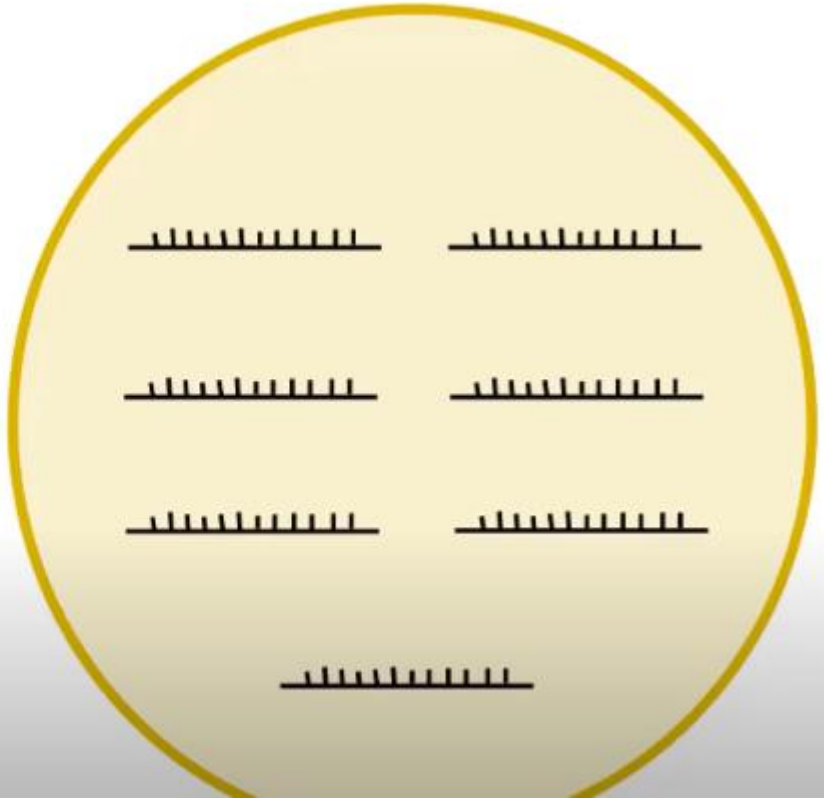


Key Spatial Transcriptomics Technologies

In-Situ Hybridization (ISH)

RNA species of interest

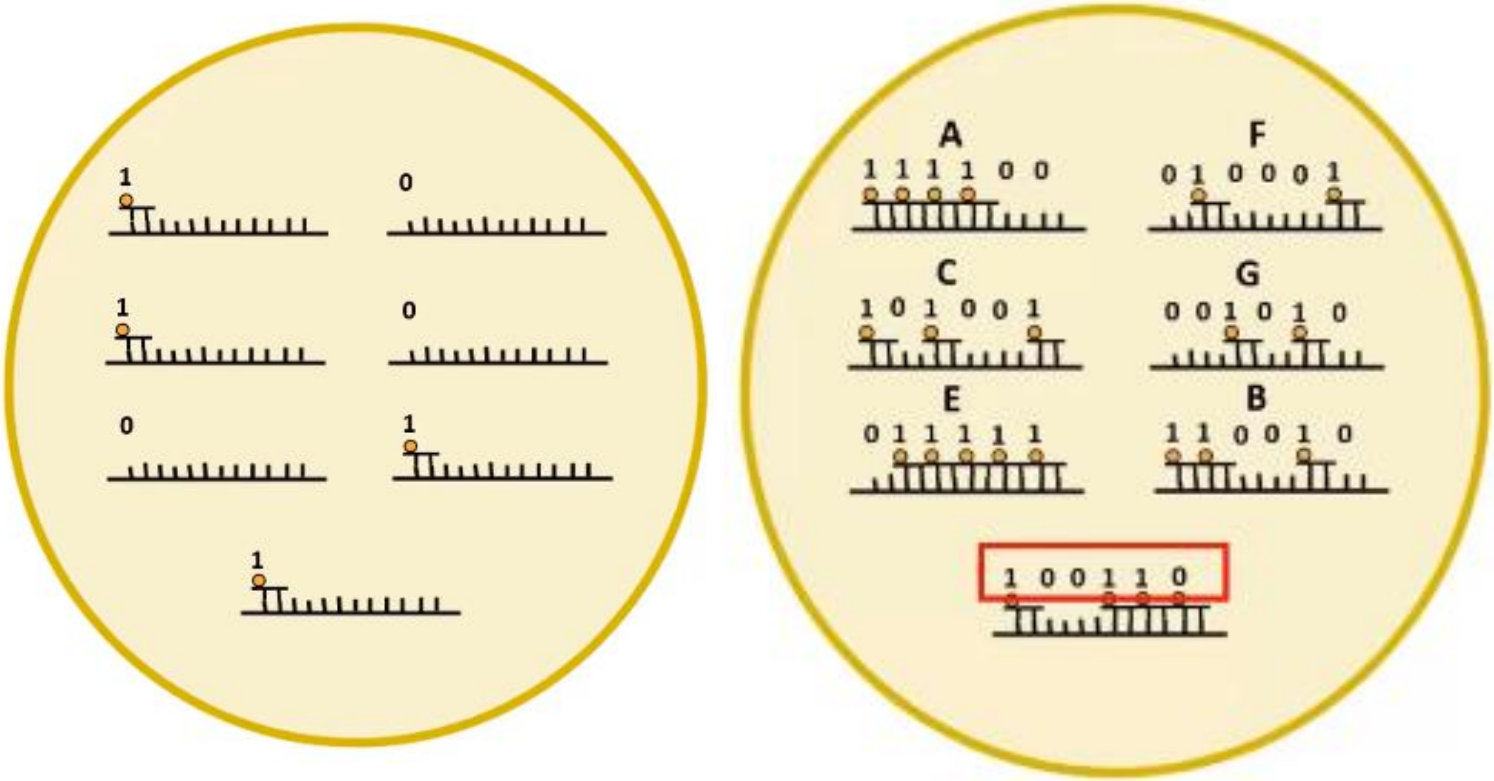
A	-	111100
B	-	110010
C	-	101001
D	-	100111
E	-	011111
F	-	010001
G	-	001010



Key Spatial Transcriptomics Technologies

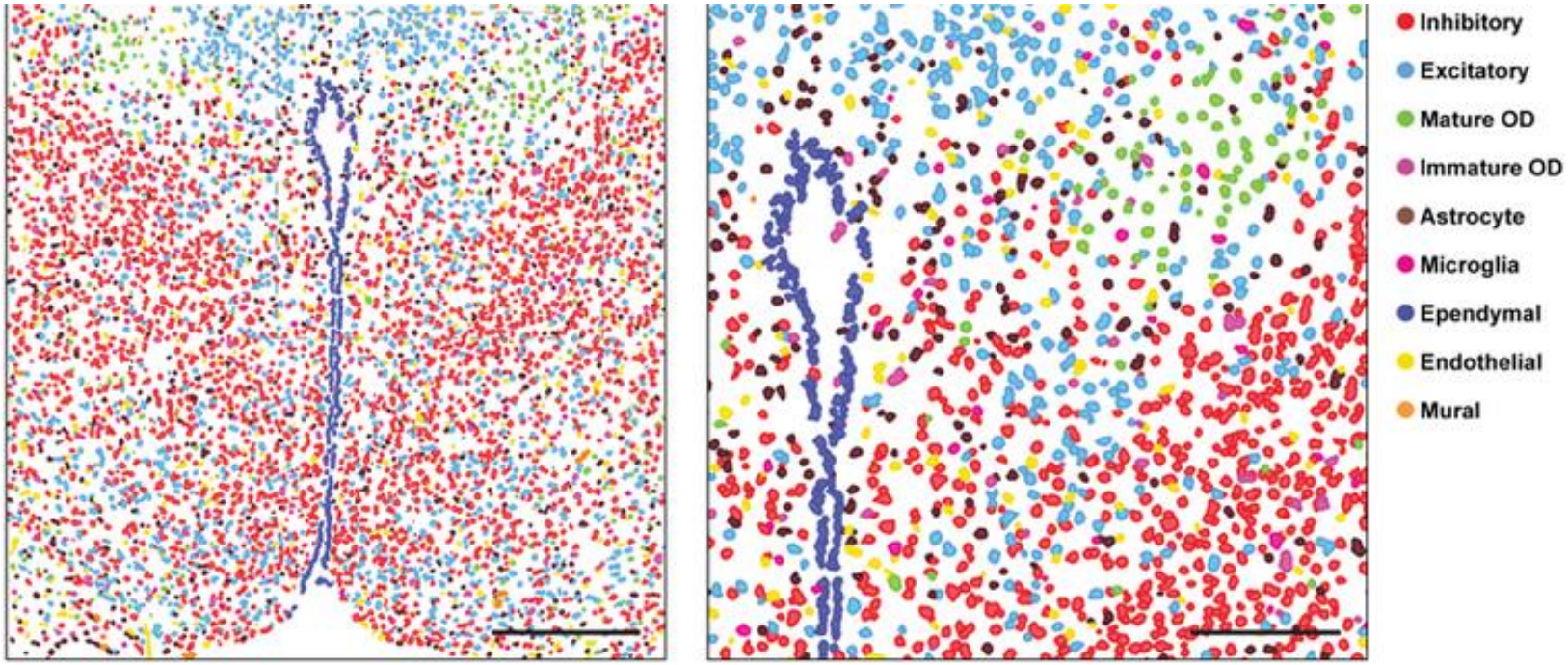
In-Situ Hybridization (ISH)

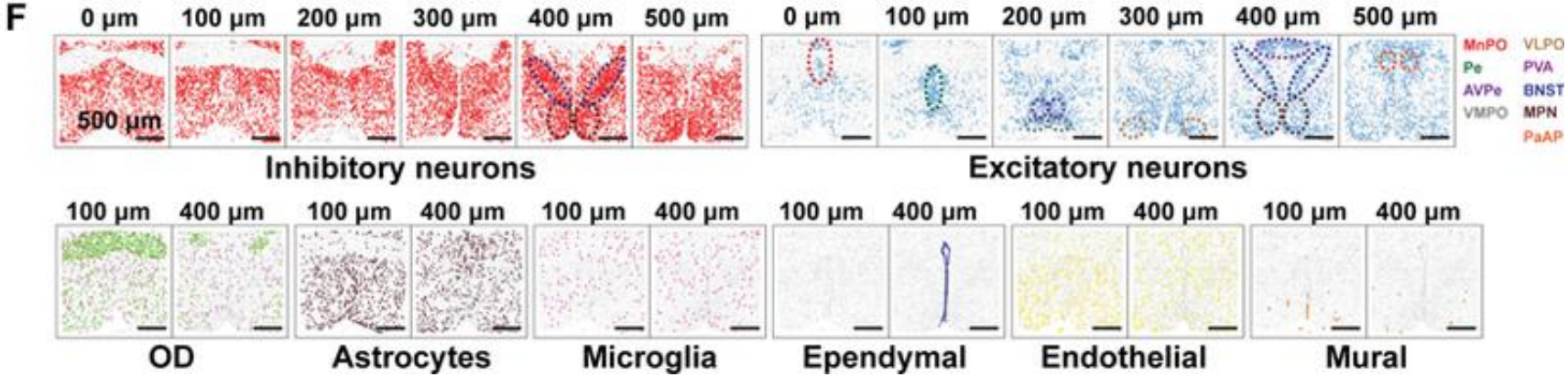
A 111100
B 110010
C 101001
D 100111
E 011111
F 010001
G 001010



Key Spatial Transcriptomics Technologies

In-Situ Hybridization (ISH)





Key Spatial Transcriptomics Technologies

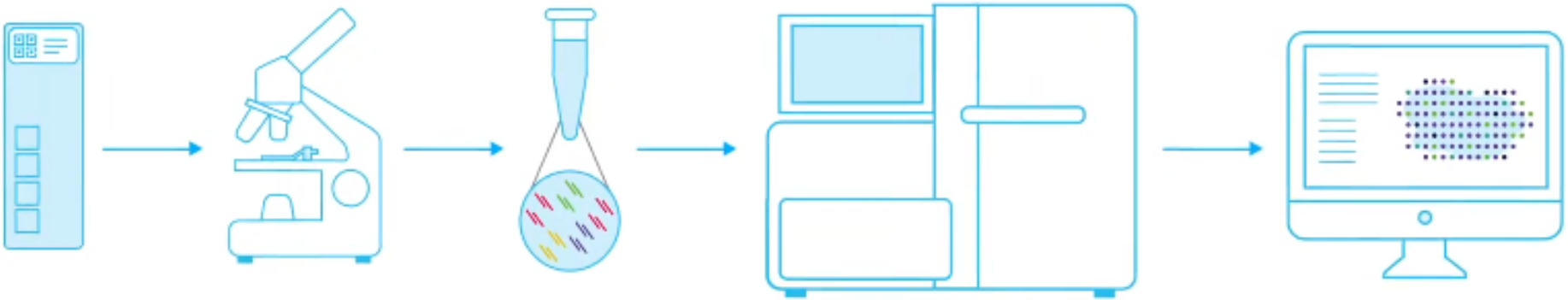
In-Situ Hybridization (ISH)

Animal_ID	Animal_sex	Behavior	Bregma	Centroid_X	Centroid_Y	Cell_class	Neuron_cluster_ID	Ace2	Adora2a	Aldh1l1	Amigo2	Ano3
1	Female	Naive	0.26	-3211.562145	2608.541476	Astrocyte	NaN	0.000000	1.638275	21.299750	0.000000	1.638275
1	Female	Naive	0.26	-3207.923151	2621.795437	Inhibitory	I-5	0.000000	0.000000	1.578873	0.000000	0.789452
1	Female	Naive	0.26	-3209.578004	2633.153494	Inhibitory	I-6	0.000000	0.000000	2.701349	5.402654	0.000000
1	Female	Naive	0.26	-3203.853515	2756.045983	Inhibitory	I-5	0.000000	0.000000	1.845116	0.922560	0.000000
1	Female	Naive	0.26	-3202.682705	2608.803635	Inhibitory	I-9	0.000000	0.000000	6.352415	0.000000	0.000000
1	Female	Naive	0.26	-3203.466059	2743.162728	OD Mature 2	NaN	0.000000	0.000000	0.000000	0.000000	0.000000
1	Female	Naive	0.26	-3195.694211	2718.529698	Inhibitory	I-5	0.000000	0.000000	1.307013	0.000000	0.000000
1	Female	Naive	0.26	-3197.408909	2641.390236	Inhibitory	I-26	0.518518	0.000000	0.000000	5.185289	0.000000
1	Female	Naive	0.26	-3196.239607	2684.324609	Inhibitory	I-35	0.000000	0.000000	0.000000	0.000000	0.000000
1	Female	Naive	0.26	-3194.392325	2743.627336	OD Mature 2	NaN	0.000000	0.000000	0.000000	0.000000	0.000000

Key Spatial Transcriptomics Technologies

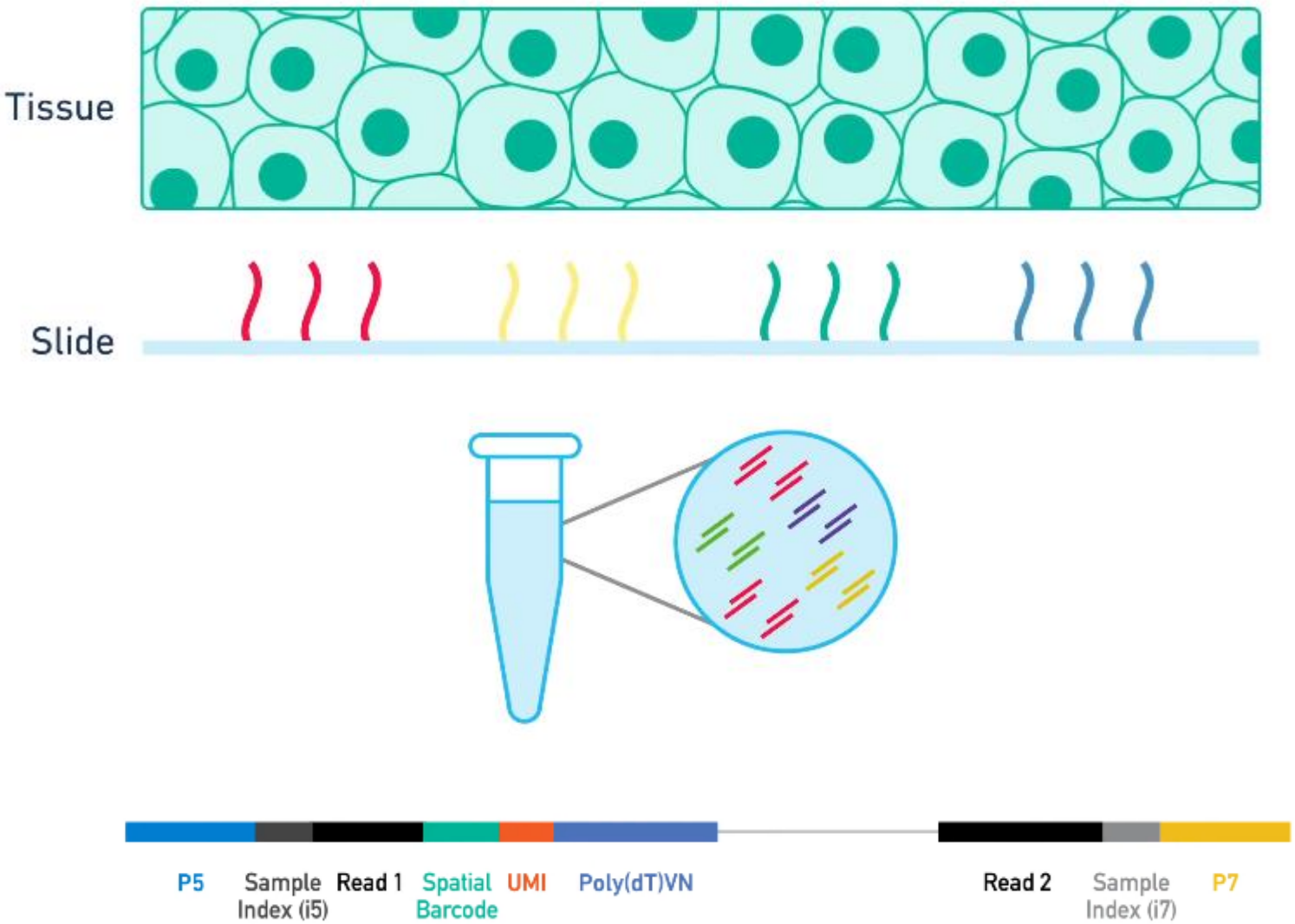
n-Situ Capturing (ISC)

- Spatial barcoding followed by sequencing
  - Each spatial region is indexed by specific barcode
  - Example: Visium from 10x Genomics
  - Spot-based (~55 µm)



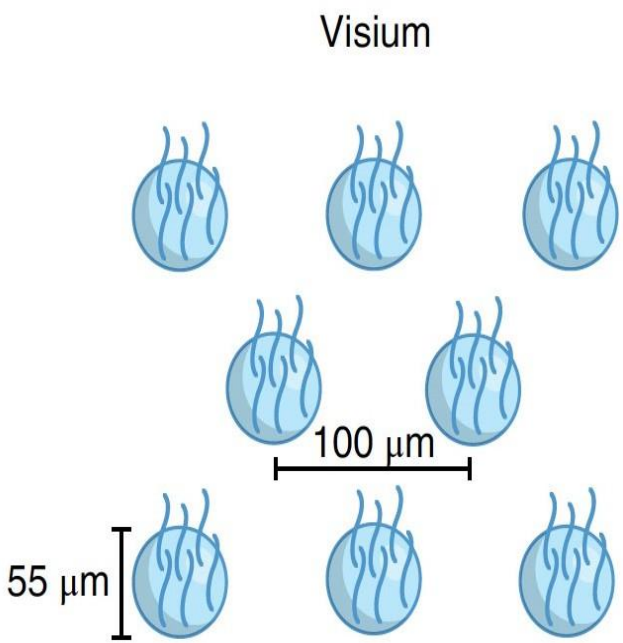
Key Spatial Transcriptomics Technologies

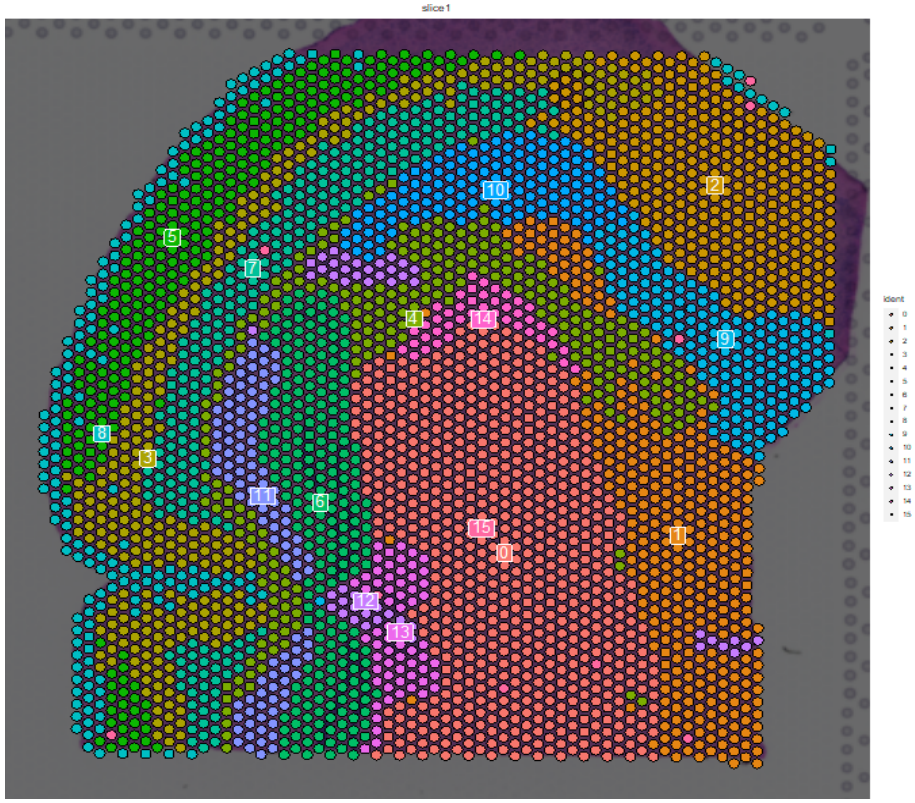
n-Situ Capturing (ISC)



Key Spatial Transcriptomics Technologies

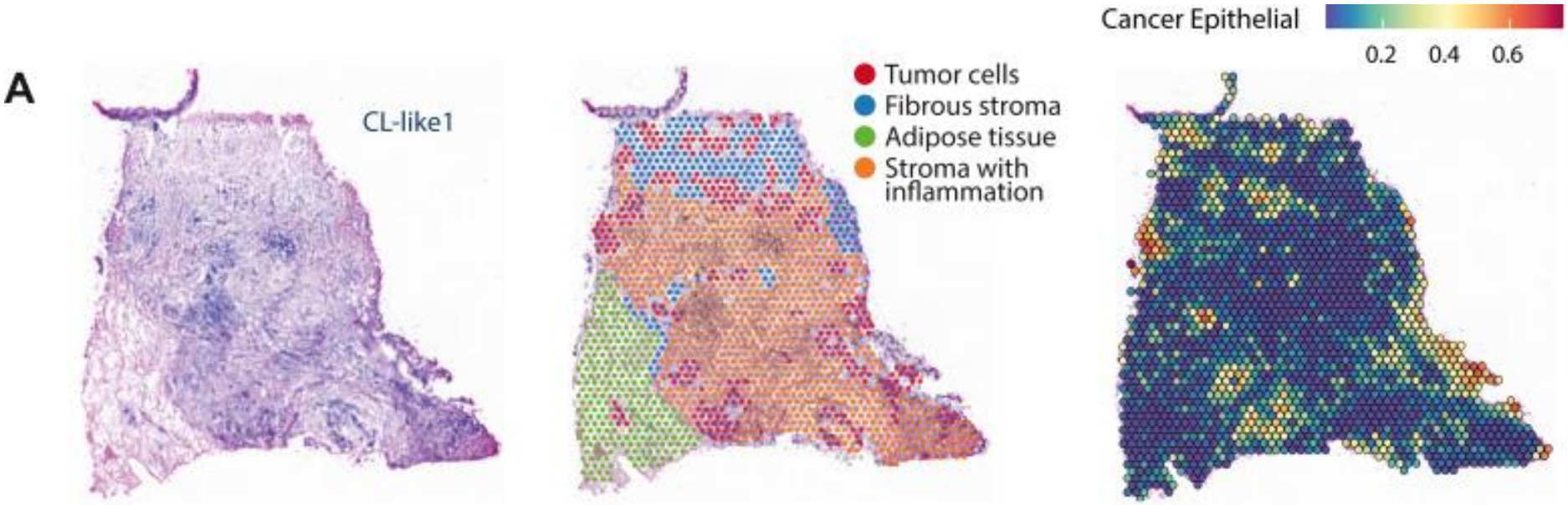
n-Situ Capturing (ISC)





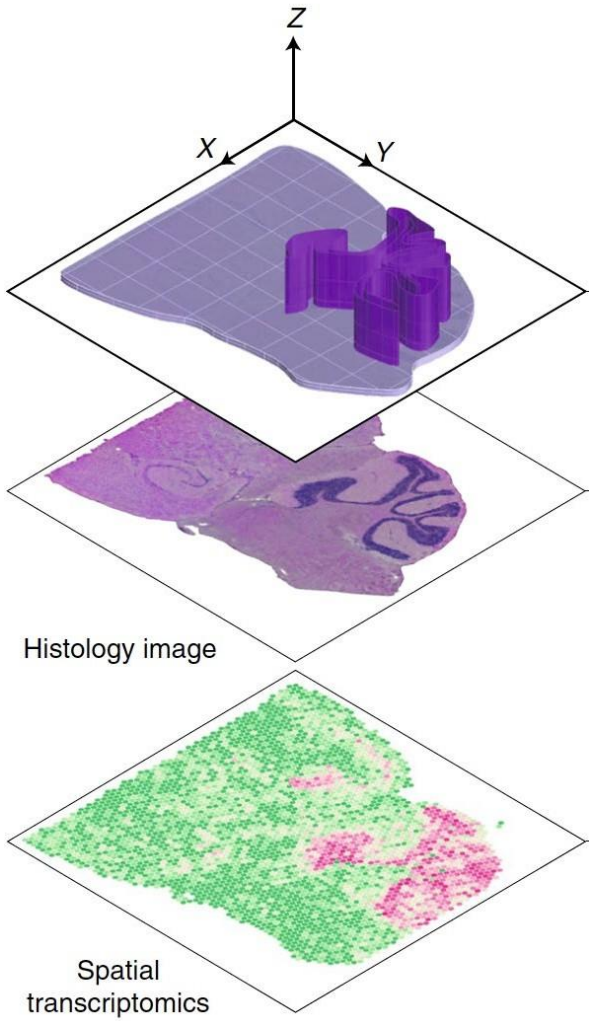
Key Spatial Transcriptomics Technologies

n-Situ Capturing (ISC)



Major Goals

- Identification of spatially variable genes
  - Spatial location
  - Histology image
  - Gene expression profile



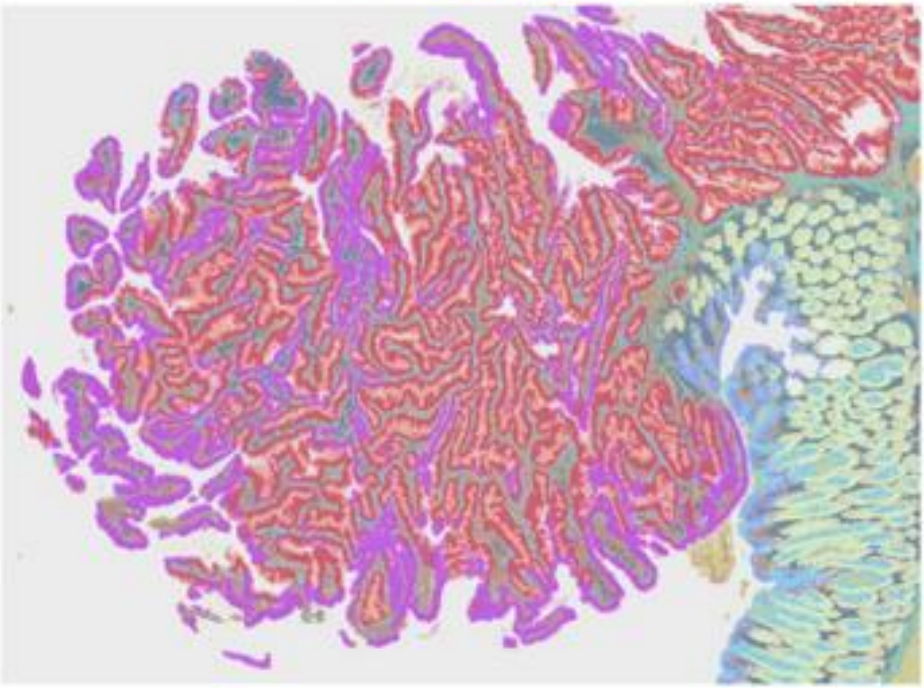
Nature Methods volume 18, pages1342–1351 (2021)

Key Spatial Transcriptomics Technologies

n-Situ Capturing (ISC)

Visium

Visium HD



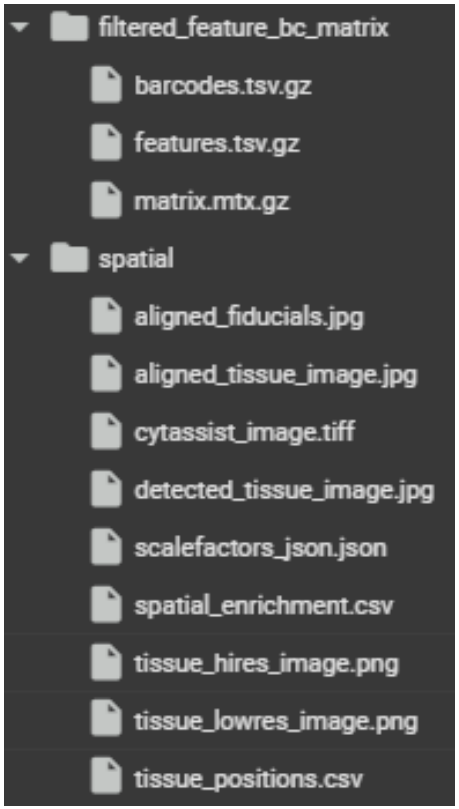
Key Parameters

- Resolution: from 50 µm to 10 nm
- Number of genes detected: from a few hundred to the entire genome
- Detection efficiency: from 5% to close to 100%
  - Molecular capture and optical crowding are limiting factors

Data Format

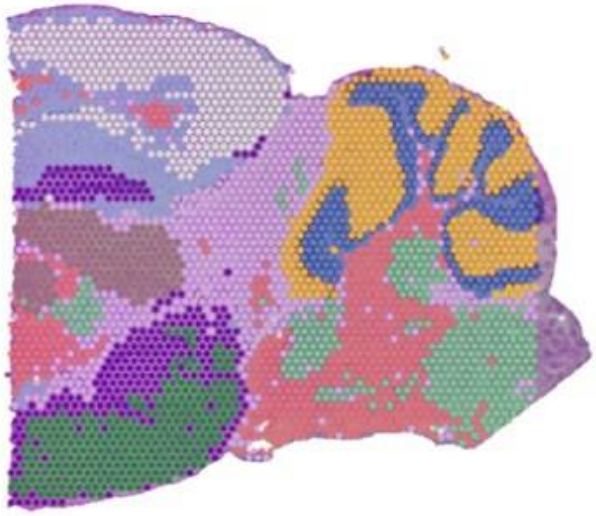
- Gene-by-location matrix

	Gene 1	*Gene 2*	*...*	*Gene m*
Spatial coordinate 1	18	0	...	3
Spatial coordinate 1	2	3	...	5
...	...	...	...	...
Spatial coordinate n	0	8	...	20



Major Goals

- Identification of spatial domains



Spatial domains

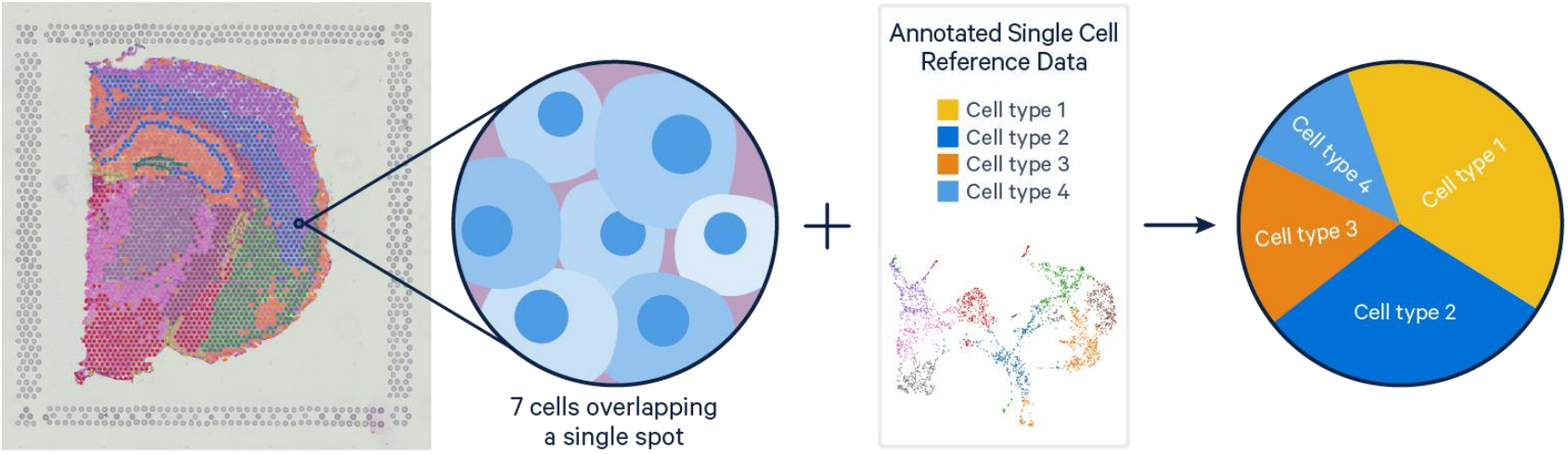
Nature Methods volume 18, pages1342–1351 (2021)

Major Goals

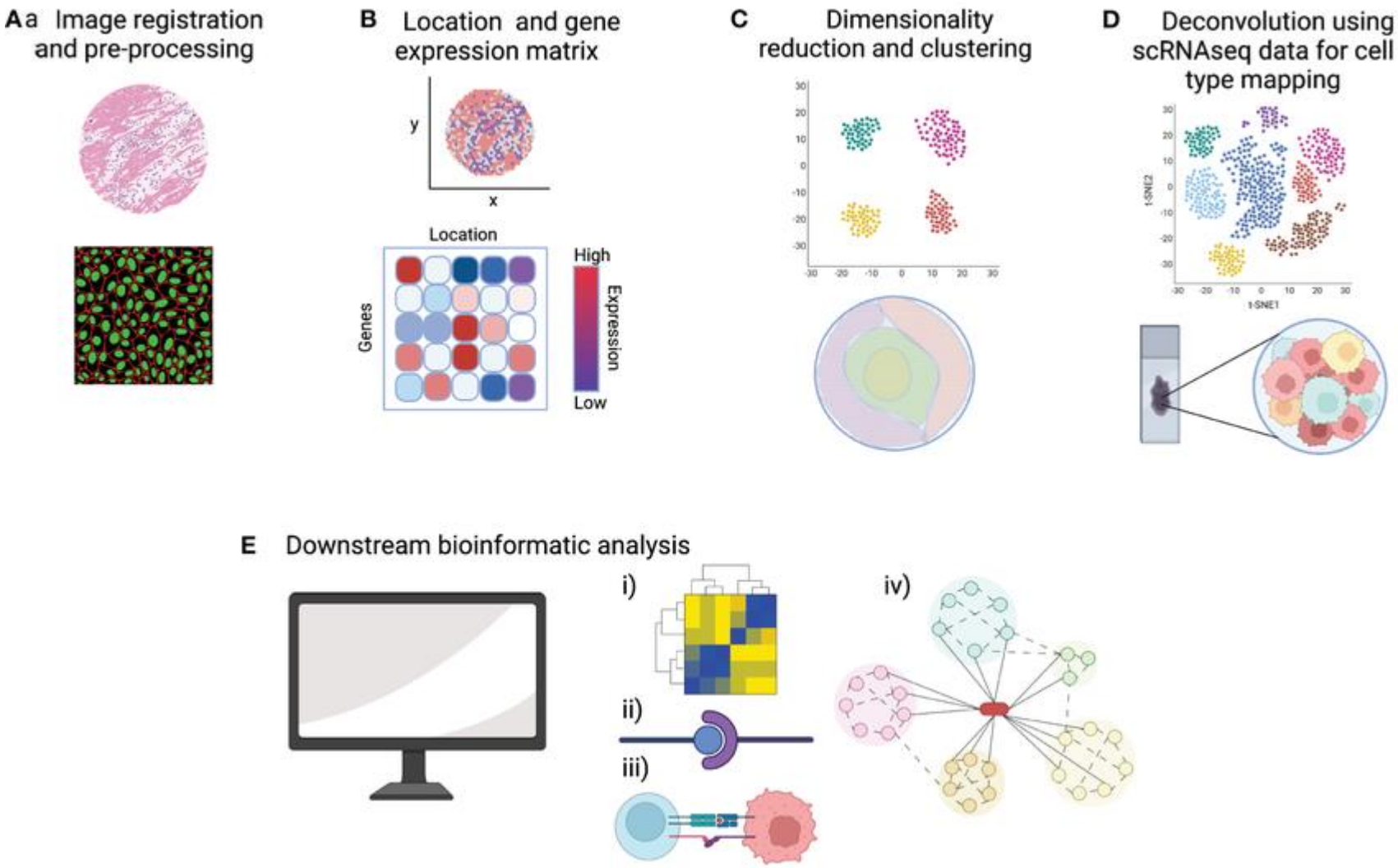
- Cell type identification (if spatial resolution can achieve single cell)
  - Based on dimensionality reduction, with the assumption that cells of the same type are similar to each other in terms of gene expression and therefore cluster together

Major Goals

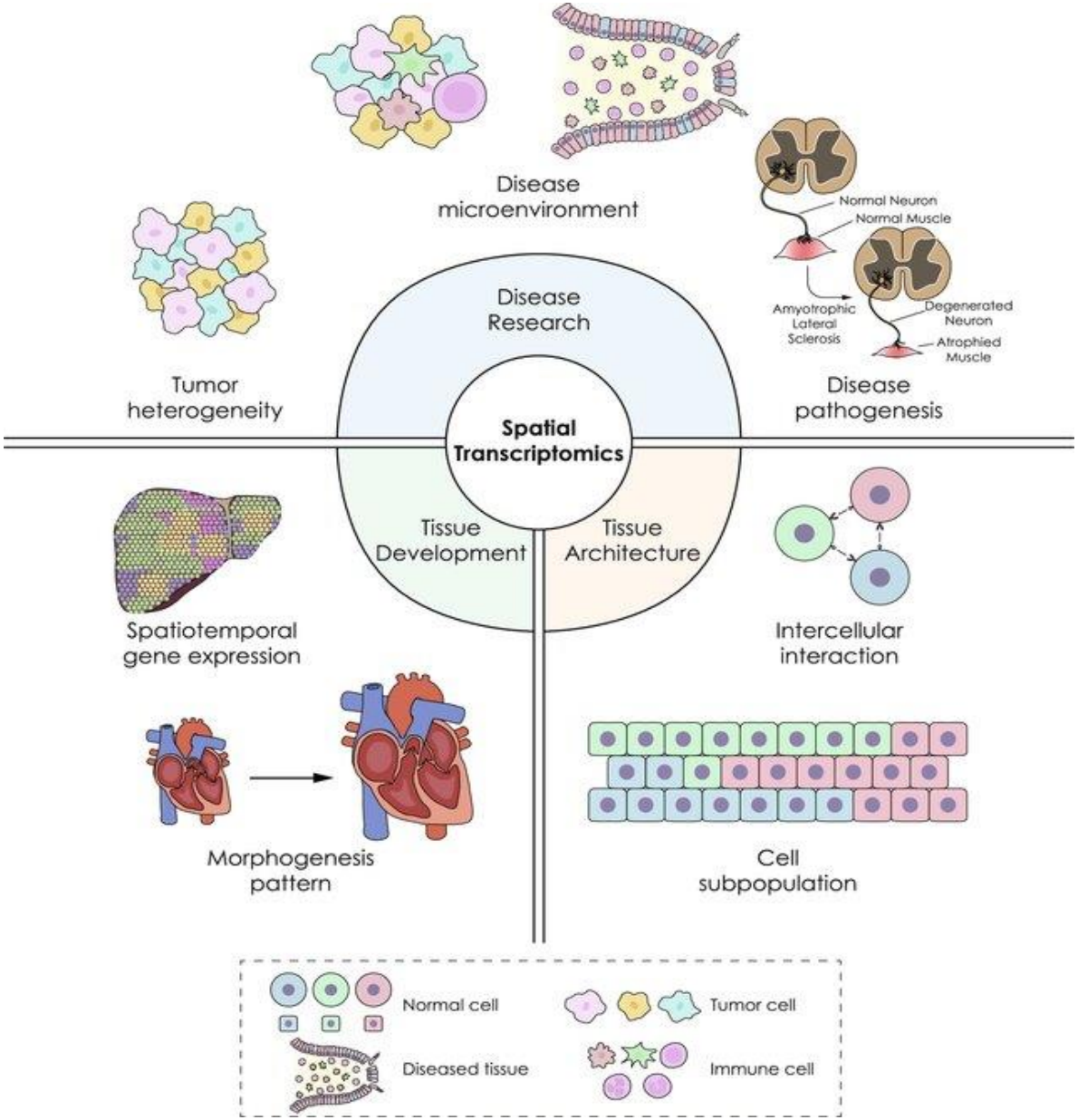
- Spatial decomposition: if spatial resolution larger than a cell, determination of what cell types are in each capture region, i.e., cellular deconvolution



Spatial Transcriptomics Data Analysis



Application of ST



Thank you! Questions?